Out-of-sample results (locked out-of-sample 2022-2025)

Frozen parameters: instrument=Nifty50, k=0.5, window=60, hold=2 days, side=both, cost=0.05% per side.
Run at 2026-11-03 15:49:24.

1. Data quality of the locked period (same cleaning function, nothing tuned)

rule	series	rows	details
muhurat session (1 hour, holiday)	nifty	3	[datetime.date(2022, 10, 24), datetime.date(2024, 11, 1), datetime.date(2025, 10, 21)]
rows kept after single-series rules	nifty	985	started with 988
muhurat session (1 hour, holiday)	bank	3	[datetime.date(2022, 10, 24), datetime.date(2024, 11, 1), datetime.date(2025, 10, 21)]
rows kept after single-series rules	bank	984	started with 987
date only in nifty (dropped)	nifty	1	[datetime.date(2025, 2, 1)]

Clean table: 984 rows, 2022-01-03 to 2025-12-31.
Days with a move above 5% in either index: 2.

2. In-sample vs out-of-sample, same rule, same parameters

	In-sample 2012-21	Out-of-sample	Buy & hold Nifty, in-sample	Buy & hold Nifty, out-of-sample	Same rule on Bank Nifty, out-of-sample
CAGR %	20.19	9.64	14.4	10.82	-3.54
Ann vol %	15.76	12.11	17.64	14.6	14.74
Sharpe	1.25	0.82	0.85	0.78	-0.17
Max drawdown %	-16.74	-16.38	-37.77	-17.11	-37.7
Days	2444	983	2444	983	983
Trades	661	263	nan	nan	279
Win rate %	51.89	49.05	nan	nan	44.09
Avg win %	1.55	1.3	nan	nan	1.38
Avg loss %	-1.07	-0.96	nan	nan	-1.15

Out-of-sample gross Sharpe (before costs) 1.38, net 0.82. Costs paid: 26.3% of capital. Time in market: 73% of days.

	2022	2023	2024	2025
strategy net %	16.4	1	4.9	16.1
buy & hold %	4.2	19.8	8.8	9.9

Alpha / beta vs always-long Nifty, out-of-sample: average exposure 0.20, beta 0.05, alpha 9.3% a year (t=1.37).

3. Is the underlying anomaly still there? (yesterday's gap -> next tradable day)

period	index	effect bps per 1sd gap	t-stat	n
research 2012-21	Nifty50	12.84	3.91	2442
research 2012-21	BankNifty	18.9	3.05	2442
out-of-sample	Nifty50	-0.18	-0.04	981
out-of-sample	BankNifty	-5.25	-0.88	981

Out-of-sample mean next tradable day return (bps) by gap quintile:

	Q1 (big gap down)	Q2	Q3	Q4	Q5 (big gap up)
bps	-5	6.2	10.3	3.5	4.2

4. Parameter sensitivity out-of-sample (information only, nothing is re-tuned from this)

variant	k	window	hold	net Sharpe	CAGR %	MaxDD %	trades
base	0.5	60	2	0.82	9.64	-16.38	263
k -20%	0.4	60	2	0.44	4.88	-15.66	286
k +20%	0.6	60	2	0.52	5.5	-19.32	250
window -20%	0.5	48	2	0.68	7.95	-21.55	268
window +20%	0.5	72	2	0.66	7.53	-15.05	255
hold -1	0.5	60	1	-0.21	-2.66	-24.26	336
hold +1	0.5	60	3	0.78	9.53	-18.49	243

Cost sensitivity out-of-sample:

cost per side %	net Sharpe	CAGR %
0	1.38	17.29
0.02	1.1	13.4
0.05	0.82	9.64
0.08	0.54	6.01
0.1	0.26	2.49

5. Was the out-of-sample drawdown within what the research-period bootstrap predicted?

Actual out-of-sample max drawdown: -16.4% over 983 days.

- Scenario 1 (edge as measured): median predicted max drawdown -14.9%, worst 5% -24.3%. Share of simulated paths with a drawdown at least as bad as the actual one: 36%.
- Scenario 2 (half the edge): median -19.7%, worst 5% -34.1%. Share at least as bad as the actual one: 72%.

Actual out-of-sample Sharpe 0.82. Share of simulated paths with a Sharpe at least this LOW: scenario 1 16%, scenario 2 66%.

Bootstrap 95% confidence interval for the out-of-sample Sharpe (resampling the out-of-sample days): -0.22 to 1.83 (median 0.82).